

MASTER'S COMPREHENSIVE EXAMINATION

Instructions: Attempt any three out of the following four questions. Indicate on your blue books which three you would like to have graded. You may use books and notes.

1. Does switching from ice cream to frozen yogurt for desserts reduce cholesterol? A study consisting of five randomly-chosen individuals yields the following cholesterol counts:

Individual	Ice Cream	Frozen Yogurt
1	210	212
2	217	220
3	209	210
4	215	213
5	204	200

Assume joint normality of the counts.

- Test the hypothesis that the mean cholesterol level is the same for those who eat ice cream as for those who eat frozen yogurt, at level .05. Find the p -value.
- Calculate a 95% confidence interval for the difference between the mean cholesterol level for those who eat ice cream and the mean for those who eat frozen yogurt.

2. Many college graduates have taken out loans to pay for college. A researcher seeks to study the loan burden on students who graduated college in the United States in May 2008. The researcher seeks to estimate the average (mean) outstanding loan amount per college graduate. The researcher plans to take a simple random sample of 150 college graduates, from all U.S. May college graduates. (You can assume that the number of graduates is very, very large.) A simple random sample of 150 students is drawn, and among them 100 students have some outstanding loan. Among the 100 students with some outstanding loan, the sample mean loan outstanding is \$2,500, and the sample standard deviation of loan outstanding is \$700.

- Give a 95% confidence interval for the proportion of all students who have some outstanding loans.
- Give a 95% confidence for the mean outstanding loan for the population of all students. (This population includes students without outstanding loans and students with outstanding loans.)

3. A regression model is fit, modeling the volume V of lumber in a type of tree as a function of the tree's diameter D . Diagnostic plots indicated that a linear relationship proved adequate. Fitting this model yielded the following coefficients, along with their standard errors

Intercept	105	5.42
D'	12.1	3.7

Here the variable D' is calculated from D by subtracting off the sample mean of D . The estimate of the residual standard deviation of V , after accounting for D' , is 2.3. These calculations were performed on a sample of 20 trees.

- Test the null hypothesis that tree diameter is unrelated to lumber volume, vs. the alternative that some relationship (positive or negative) exists. Use a test of size .05. Report a p -value.

- b) Estimate the mean volume of trees whose diameter matches the mean of the sampled trees' diameters. Construct an interval in such a way that such intervals contain the population mean value 95% of the time.
- c) Predict the volume of a new tree whose diameter matches the mean of the sampled trees' diameters. Construct an interval in such a way that such intervals contain the new value 95% of the time.

4. Below is an ANOVA table as for a factorial design. The ANOVA table implicitly incorporates the assumption that all effects are fixed. In fact, although A is fixed, B is a random effect, so the values of the F statistic are incorrect.

Source	df	SS	MS	F
A	3	7410	2470	123.5
B	5	800	160	8.0
A×B	15	1050	70	3.5
Error	216	4320	20	

For the questions below, you should use the unrestricted mixed model.

- a) Please show the expected mean square for each row of the table.
- b) Please calculate the correct value for the F statistic for each term as well as the corresponding degrees of freedom.
- c) Please estimate σ_B^2 and $\sigma_{A \times B}^2$